Human Activity Recognition and Person Identification system from Wi-Fi signals with Contrastive Learning

Nicolò Spuri, Alessio Marchetti, Sirio Trentin†

***Abstract*—Human Activity Recognition (HAR) and Person Identification (PI) are fundamental components for next-generation smart environments, security surveillance, and personalized user systems. While traditional vision-based solutions raise severe privacy concerns, passive Wi-Fi sensing has emerged as a highly effective, non-intrusive alternative. In this paper, we present two robust Deep Learning architectures that leverage the Doppler shifts in Wi-Fi signals to accurately perform both HAR and PI. To overcome the critical challenge of generalizing across different subjects, time spans, and environmental layouts, we implement advanced representation learning techniques on a comprehensive dataset of Doppler traces. Specifically, we propose two distinct pipelines: a self-supervised contrastive approach that utilizes the signals from multiple receiver antennas as naturally augmented views, and a supervised contrastive framework optimized for task-specific feature clustering. Both methodologies demonstrate exceptional domain generalization and efficiency. The self-supervised model achieves an impressive average accuracy of 100% on Activity Recognition and 97% on Person Identification. Similarly, the supervised model, which is strictly dedicated to Activity Recognition, reaches a 97% average accuracy. These results strongly validate the efficacy of contrastive pre-training in building resilient, data-efficient, and environment-independent Wi-Fi sensing systems.**

***Index Terms*—Human Activity Recognition, Person Identification, Wi-Fi Sensing, Self-Supervised Learning, Contrastive Learning.**

I. Introduction

Human Activity Recognition (HAR) and Person Identification (PI) are critical components of smart environments, with applications ranging from elderly care and security surveillance to personalized user experiences. Traditional vision-based systems, while effective, raise significant privacy concerns and are sensitive to lighting conditions. This has pushed research into alternative, non-intrusive sensing modalities.

Wi-Fi based sensing has emerged as a promising passive and privacy-preserving alternative. It leverages the fact that human movements disturb the propagation of radio frequency (RF) signals between transmitter and receiver [1]. These disturbances are captured in the Channel State Information (CSI), a fine-grainged measurement of the channel's properties across multiple subcarriers. By analizing the temporal variations in CSI, specifically through the Doppler spectrum, it is possible to infer the presence and nature of movements in the environment.

Despite its potential, Wi-Fi based sensing faces significant challenges. The performance of machine learning models trained on CSI data often degrades when deployed in a new environmental layout, furniture, and hardware specifics. Acquiring large, labeled datasets that cover this vast domain diversity is expensive and time-consuming.

To address these challenges, we propose two pipelines based on self-supervised [2] and supervised contrastive learning [3]. Our key insight is that the multiple antennas on a commodity Wi-Fi device provide naturally occurring, diverse views of the same physical event, furthermore we can exploit what we know from the labels of the data samples.

- In the first pipeline we treat the Doppler spectra from our receiver antennas as four positive views of a single data sample. A deep neural network is pre-trained on a large amount of unlabeled data to learn representations that are invariant to the specific antenna view, this is then fine-tuned on a smaller labeled dataset for the downstream tasks of Activity Recognition (AR) and Person Identification (PI) using a two-headed architecture.
- In the second pipeline we treat samples that share the same label as positive keys. A deep neural network is pre-trained on a large amount of unlabeled data to generalize among different environments, moments and persons, this is then fine-tuned on a smaller labeled dataset using a single head concentrating only on the AR task.

Our approach demonstrates that contrastive pre-training can effectively learn robust features, paving the way for more generalizable and data-efficient Wi-fi sensing systems.

The main contributions and novelties of this work are manifold:

- We introduce a novel approach to Wi-Fi sensing that naturally leverages the multiple antennas of a access point, enabling highly effective self-supervised contrastive pre-training.
- Our self-supervised framework successfully implements a multi-head architecture, allowing the system to perform both Activity Recognition and Person Identification simultaneously.
- We demonstrate that integrating a supervised contrastive learning framework significantly enhances the robustness

†Department of Information Engineering (DEI), University of Padova, 35131 Padova, Italy,
E-mail: {nicolo.spuri, alessio.marchetti, sirio.trentin}@studenti.unipd.it

of the AR task.

These advancements open up numerous practical applications. A robust, non-intrusive AR and PI system can be seamlessly integrated into smart home environments to provide personalized automation (e.g., adjusting lighting or temperature based on who is in the room and what they are doing). Furthermore, it holds significant potential for privacy-preserving security surveillance and elderly care monitoring, where detecting specific movements or falls without the use of cameras is highly desirable.

While our models demonstrate high efficiency, there is room for future improvements. Subsequent work could focus on extending the framework to handle complex multi-subject scenarios where multiple people perform different activities concurrently. Additionally, expanding the number of recognized activities and optimizing the network architecture for real-time inference on edge devices would further enhance the system's real-world deployability.

The remainder of this paper is organized as follows: Section II reviews the related work concerning Wi-Fi-based sensing and contrastive learning paradigms. Section III outlines our two proposed processing pipelines. Section IV details the data acquisition, signal processing, and feature extraction steps. Section V describes the underlying self-supervised and supervised learning frameworks. Section VI presents the experimental results and evaluates the performance of the proposed models. Finally, Section VII draws the conclusions and discusses future research directions.

II. Related Work

In recent years, Wi-Fi-based Human Activity Recognition (HAR) has gained significant traction as a non-intrusive alternative to vision-based monitoring. While many existing Wi-Fi HAR systems rely on active sensing—requiring the subject to carry a Wi-Fi-connected device—passive approaches have emerged as a more practical solution. Our research focuses on this passive paradigm and is heavily inspired by recent advancements in signal processing and representation learning.

A. Passive Wi-Fi Sensing and the SHARP Framework

Our models are built upon the foundational work of the SHARP (Environment and Person Independent Activity Recognition) system introduced by Meneghello et al. [1]. SHARP demonstrated that passive sensing using commodity IEEE 802.11 access points can achieve highly accurate activity recognition, boasting an average accuracy of 95%. The system achieves this by transforming raw Channel State Information (CSI) into Doppler spectrograms, effectively mapping how human movement alters the surrounding Radio Frequency (RF) environment.

Furthermore, our models are trained and evaluated using the comprehensive dataset published alongside the SHARP system [4], [5]. This dataset utilizes an 80 MHz Wi-Fi channel and a four-antenna configuration to capture high-resolution measurements. Crucially, the data undergoes a rigorous extraction and sanitization process to remove hardware-induced phase offsets and anomalies, resulting in the clean Doppler traces that serve as the input for our neural networks. While SHARP represents a state-of-the-art solution, its architecture is exclusively tailored for the Activity Recognition (AR) task. This single-task focus limits its deployment in more complex smart environments where user personalization—such as Person Identification (PI)—is also required.

B. Contrastive Representation Learning

To overcome the limitations of SHARP and adapt the system for both AR and PI, our work integrates Contrastive Learning paradigms. As comprehensively reviewed by Le-Khac et al. [2], contrastive representation learning is a powerful framework that teaches a model to distinguish between similar and dissimilar data points. By pulling similar samples (positive pairs) together and pushing dissimilar samples (negative pairs) apart in the embedding space, the network learns rich, generalized features without necessarily relying on vast amounts of labeled data. This approach is highly effective in making models robust to the environmental and temporal changes that typically degrade Wi-Fi sensing systems.

C. Self-Supervised and Supervised Contrastive Approaches

In this paper, we employ two distinct strategies to generate positive pairs for the contrastive loss function. The first strategy utilizes a Self-Supervised Contrastive approach, which relies on domain-specific data augmentation applied across the signals received by the four different antennas, teaching the model to recognize the same physical event regardless of the specific antenna's "view."

The second strategy leverages the Supervised Contrastive Learning (SupCon) framework introduced by Khosla et al. [3]. Khosla et al. demonstrated that traditional contrastive learning could be significantly improved by incorporating label information during the pre-training phase. Instead of relying solely on augmented versions of a single sample, SupCon treats entirely different samples belonging to the same class as positive keys. Their work proved that this supervised approach yields representations that are remarkably robust to noise, outperforming standard cross-entropy training. By adapting this framework to Doppler traces, we force the network to cluster samples of the same activity closer together in the latent space, regardless of the person performing it or the environment they are in.

D. Architectural Foundations: ResNet-34 and SimCLR

To effectively process the complex temporal and spatial information captured in the Doppler traces, our models draw upon established deep learning architectures and self-supervised frameworks. Specifically, we utilize ResNet-34 [6] as the core backbone for feature extraction in our self-supervised approach to extract complex hierarchical features from the spectrograms while mitigating the vanishing gradient problem.

Furthermore, our multi-antenna pre-training strategy is heavily inspired by SimCLR [7].We adapt its foundational NT-Xent loss for the RF domain. Instead of just two augmented views, we modify the loss function into a multi-positive formulation that simultaneously leverages the signals from all four receiver antennas as naturally occurring positive views of the same physical event.

By combining the sanitized, high-quality CSI dataset from [4], [5] with the advanced representation learning techniques detailed in [2] and [3], our proposed pipelines successfully extend the capabilities of traditional Wi-Fi sensing to handle both generalized Activity Recognition and specific Person Identification.

III. Processing Pipeline

The proposed approach aims at exploiting the temporal and spatial information in the Doppler traces obtained from the raw WiFi signals in order to perform an activity recognition and person identification task.

First the raw Doppler spectrum are collected from multiple antennas and processed to generate the input samples for our neural network. The CSI measurements associated with the different antennas are grouped together and each activity sequence is segmented into temporal windows of fixed length. Then the dataset is split into training, validation and test set, ensuring that the model is evaluated on different portions of the activity sequences while avoiding excessive overlapping between the samples. We then implemented two neural networks, one using a self-supervised approach for activity recognition and person identification, while the other using a supervised contrastive framework in order to be more robust on the activity recognition task.

In the first approach, data augmentation is applied to the signal windows through random transformations performed independently on each antenna. Samples originating from the same window, but captured by different antennas, are then treated as positive keys by the NT-Xent loss function, which is adapted to accommodate more than two positive keys. Consequently, in the initial phase, a neural network consisting of a backbone and a head undergoes self-supervised contrastive pretraining. The backbone learns to extract the signal features, deriving an optimal representation for each environment-person-activity association. Finally, this pre-trained backbone is coupled with two distinct heads: one for PI and one for AR.

The pipeline of this approach is illustrated in Figure 1

The second approach instead is divided in different stages: the first consist of a supervised contrastive pre-training phase. During this step, the network learns a compact representation of the Doppler traces by using the activity label to bring samples belonging to the same class closer in the embedding space. After this step, the feature extractor is transferred to a classification network, the parameters of the pre-training are frozen and only the final classification layer is optimized. By doing that the final model gives in output the predicted activity, while being very robust and reducing the risk of overfitting.

The complete pipeline of this second approach is illustrated in Figure 2.

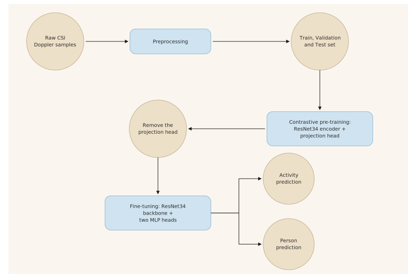

Fig. 1: Pipeline of the self-supervised contrastive approach

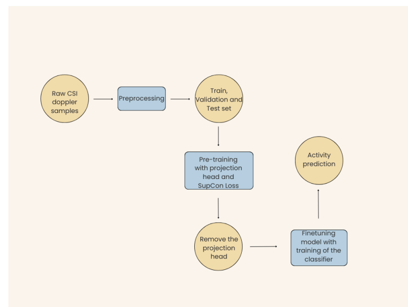

Fig. 2: Pipeline of the supervised contrastive approach

IV. Signals and Features

In this work, we employ a preprocessed dataset of Doppler traces [4], [5]. The original data collection relies on two strategically positioned monitoring devices to capture reliable Doppler shift estimates. Specifically, the underlying Wi-Fi infrastructure utilizes Orthogonal Frequency Division Multiplexing (OFDM) to transmit data across K partially overlapping, orthogonal sub-channels. At the receiver, amplitude and phase are estimated for each sub-channel and stored in a Channel State Information (CSI) matrix. This complex matrix characterizes the Channel Frequency Response (CFR) along every receiving antenna. Subsequently, a sanitization step is executed to filter hardware-induced phase offsets. This procedure is strictly necessary, as these inherent anomalies affecting standard Wi-Fi transmissions would otherwise overwhelm and obscure the subtle variations required for the analysis.

Each Doppler vector is obtained by applying a Fourier transform to a short block of consecutive channel estimates, and describes how the reflected power is distributed over the velocities of the scatterers in the scene. The transform is evaluated on 100 points, which sets the number of velocity bins covering the unambiguous velocity range of about +/- 5 m/s. A new Doppler vector is computed for every received packet, and 340 consecutive vectors, corresponding to roughly two seconds of channel observation, are stacked along the time axis. Each sample provided to the network is therefore a 100x340 Doppler spectrogram, in which the vertical axis represents velocity and the horizontal axis represents time [1]. The spectrograms of the various possible activities included in the dataset are shown in Fig. 3.

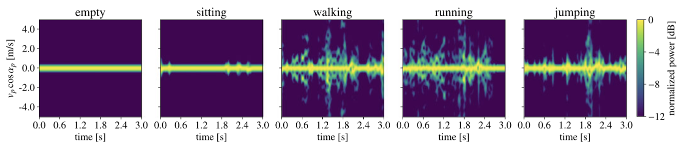

Fig. 3: Doppler spectrograms for different cases [1]

The dataset is structured into seven distinct sets, each containing various combinations of subjects, environments, and activities. In the first approach, the data from these sets were partitioned into 60% for training, 20% for validation, and 20% for testing, with the exception of Set 6. This latter set was strictly reserved as a hold-out set, remaining completely unseen by the network. During the self-supervised training phase, a data augmentation pipeline is applied independently to the Doppler spectrogram of each antenna. Specifically, following an initial per-sample standardization (yielding zero mean and unit variance), the signals are subjected to a stochastic sequence of transformations. These operations include circular time shifting, temporal and frequency masking on the Doppler axis, random amplitude scaling, additive Gaussian noise injection, and random patch erasing.

In the second approach, we used only the first set as training set (still dividing it in 60% training, 20% validation and 20% testing) while the sets from 2 to 7 are used for the testing. Furthermore, in this second approach no transformation is applied to the Doppler spectrogram.

V. Learning Framework

The first framework employs a two-stage training strategy for the network. Initially, a self-supervised pre-training phase is conducted to enable the network to learn optimal embeddings for the input signals. Subsequently, a fine-tuning phase is performed on two dedicated heads: one for Person Identification (PI) and another for Activity Recognition (AR).

A. Self-Supervised Contrastive approach architecture

As the network backbone, both an Inception architecture, similar to the one employed in SHARP [1], and a ResNet-34 [6] were evaluated. The former, however, yielded sub-optimal results and was subsequently discarded, likely due to its insufficient network depth. Conversely, the ResNet-34 architecture demonstrated superior performance, as its deeper structure and residual connections facilitated the extraction of more complex hierarchical features from the spectrograms while mitigating the vanishing gradient problem.

B. Phase 1: self-supervised contrastive pre-training

No label is used in this stage: each window is a distinct instance and its four antennas act as *natural views* of the same event, each augmented independently with the pipeline described above. A temporary projection head maps the backbone features into a 128-dimensional L_2-normalized space, where the inner product coincides with the cosine similarity, and is discarded once pre-training is complete.

The starting point is the NT-Xent loss of SimCLR [7], which assumes two views per instance and therefore a single positive per anchor. Since each window here produces $V = 4$ views, we adopt a *multi-positive* formulation: for a batch of N windows we obtain $M = N \cdot V$ embeddings and, for the anchor i, the positive set $P(i)$ collects the remaining $V - 1$ views of the same window, while every other row of the batch acts as a negative. Denoting by $\mathbf{z}_i$ the normalized projection of view i and by τ the temperature, the loss is

$$\mathcal{L} = \frac{1}{M} \sum_{i=1}^{M} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp\big(\mathbf{z}_i^\top \mathbf{z}_p / \tau\big)}{\sum\limits_{a \neq i} \exp\big(\mathbf{z}_i^\top \mathbf{z}_a / \tau\big)}, \qquad (1)$$

where the denominator runs over the whole batch except the anchor itself, whose self-similarity is masked out before the $\log$-$\mathrm{softmax}$ normalization.

This is the same multi-positive form as the Supervised Contrastive loss, but the grouping is given by the *window identity* instead of the class label, so the supervision stays entirely self-generated. Averaging the log-ratio *outside* the logarithm makes the gradient of each positive independent of $|P(i)|$, and the four antennas are exploited jointly rather than in pairs, giving three positives and $M - V$ negatives per anchor in a single forward pass. Optimization uses AdamW ($\eta = 2{\cdot}10^{-4}$, weight decay 10^{-4}) with cosine annealing down to $\eta/50$, batches of 64 windows (256 views) and $\tau = 0.15$, for 100 epochs.

C. Phase 2: two-head fine-tuning

The projection head is discarded and the pre-trained backbone is reused as a shared feature extractor for two parallel MLP heads, one for Activity Recognition (5 classes) and one for Person Identification. Both are supervised with a categorical cross-entropy and optimized jointly as

$$\mathcal{L}_{\mathrm{FT}} = \mathcal{L}_{\mathrm{CE}}(\hat{y}_{\mathrm{AR}}, y_{\mathrm{AR}}) + \lambda\, \mathcal{L}_{\mathrm{CE}}(\hat{y}_{\mathrm{PI}}, y_{\mathrm{PI}}), \qquad \lambda = 1, \quad (2)$$

so that the shared representation is forced to retain both the activity dynamics and the subject-specific gait signature; the weight λ balances the two tasks and is kept at 1, since neither objective is treated as auxiliary.

Fine-tuning proceeds in two steps. First a *linear probe*: the backbone is frozen (its batch-normalization statistics included) and only the two heads are trained for 15 epochs with AdamW ($\eta = 10^{-3}$, weight decay 10^{-3}), which measures how linearly separable the self-supervised features already are and prevents the randomly initialized heads from corrupting them. Then a *full fine-tune*: the backbone is unfrozen and the whole network is trained for at most 10 epochs with a ten times smaller learning rate ($\eta = 10^{-4}$, weight decay 10^{-4}). Model selection uses the sum of the validation accuracies of the two heads, with early stopping after 5 epochs without improvement and restoration of the best checkpoint.

A. Supervised Contrastive approach architecture

The framework of the second approach adopts a two-stage transfer learning paradigm: a supervised contrastive learning, followed by a specific fine-tuning phase for the classification task. The core of this approach is still the architecture used in SHARP [1] with the primary building block being *ReductionABlockSmall*, which utilizes pooling branches and parallel convolution in order to extract multi scale features while reducing the spacial and temporal dimensions of the input.

1) Phase 1: supervised contrastive pre-training: To extract robust representations, we employ the Supervised-Contrastive Loss (SupCon Loss)

$$\mathcal{L}_{\text{out}}^{\text{sup}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp\left(\boldsymbol{z}_i \cdot \boldsymbol{z}_p / \tau\right)}{\sum\limits_{a \in A(i)} \exp\left(\boldsymbol{z}_i \cdot \boldsymbol{z}_a / \tau\right)}. \tag{3}$$

the goal is to pull embeddings of the same activity class closer together in the latent space while pushing apart embeddings from different classes. Then a temporary *Projection Head* is attached to the backbone to map high-dimensional features into a 128-dimensional L2-normalized space, where the loss is calculated. The optimization consists in a 20 epochs training using the Adam optimizer with a learning rate of 10^{-4} and a temperature parameter $\tau = 0.07$.

2) Phase 2: Fine-Tuning for classification: Once pre-training is complete the *Projection Head* is removed and replaced by a *linear classification Head*, furthermore the weights of the backbone are freezed and only the final fully connected layer is trained. In this phase a *Categorical Cross Entropy Loss* is used to guide the training and map the extracted features into the 5 target activity classes. In the end, to prevent overfitting, a Dropout layer (0.2 rate) is used before the final one and training is governed by an *Early-Stopping* mechanism based on the validation accuracy.

3) Decision Fusion: Since the setup utilizes 4 different antennas to obtain the signals, the framework implements a *Decision Fusion* strategy where, instead of relying on a single antenna prediction, the model aggregates the output of all the antennas associated with the same time slot. The final class assignment is determined using a *Majority Voting* criteria or, in case of tie, by evaluating the sum of the predicted logits across all antennas to ensure the maximum reliability.

VI. Results

A. Self-supervised approach: results

The two-head model was evaluated on the within-domain test split (sets S1–S5, S7; windows disjoint in time from training) and on the held-out set S6, never seen during pre-training nor fine-tuning. On the within-domain split the Activity Recognition head reaches a perfect score, with 100% accuracy and unitary F1 on all five classes, while the Person Identification head attains 95.7% accuracy (F1 = 0.958): the confusion matrix shows that the residual error is almost entirely a one-directional confusion between P1 and P2, whereas P3 is never misclassified.

On the held-out set the Activity Recognition head still classifies every window correctly (100% accuracy), confirming that the multi-view contrastive pre-training extracts activity-specific Doppler features that transfer to an unseen monitor position and acquisition day. The Person Identification head labels 99.1% of the held-out windows with the correct subject; this figure, however, must be read as a consistency check rather than a generalization score, since S6 contains a single subject and therefore does not probe the discriminative ability of the head. Overall, the two objectives coexist without interference: the shared representation supports exact activity recognition while remaining informative enough about the subject, which indicates that the identity-related component of the signal is not suppressed by the augmentation pipeline.

B. Supervised contrastive approach: results

We now analize the results of the second approach, first on the intra domain test set (S1) and then in order to test the generalization capabilities on six unseen cross-domain scenarios (S2-S7).

1) Intra-domain Performance: At the individual antenna level, the fine-tuned model achieve an accuracy of 98.8%, however by applying the Decision Fusion strategy we reach a perfect classification with an accuracy of 99.9%. This demonstrates that while individual antennas may occasionally misclassify due to noisy signal, the aggregate decision across the four antennas provides an optimal prediction. In figure 4 the confusion matrices of this part.

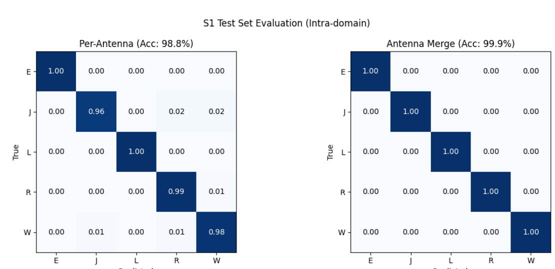

Fig. 4: Confusion matrices on the Intra-domain test set S1

2) Cross-domain Generalization: In this phase we tested to model on data from environment and subjects never seen during training:

- **Time and subject generalization:** On set S2 (different day, same room) the model maintained an accuracy of 99.89% while on S3 (both day and subject changed)

the accuracy remained at 99.19%, proving that the contrastive pre-training phase successfully extracted person-independent features.
- **Line of Sight blocked:** In scenarios where the line of Sight was blocked (S4 and S5) the performance stayed above 99%, indicating the network is capable of identifying activity specific component even when the signal path is attenuated.
- **Environmental Change:** The model showed high robustness to a new room (S6) with 98.36% accuracy. In the most challenging scenario, S7 (different environment, day, person) resulted in a drop to 89.07%. The confusion matrix for S7 in figure 5 reveal that the jumping activity is frequently confused with walking, suggesting that significant changes in room geometry and subject simultaneously can shift the Doppler distribution significantly.

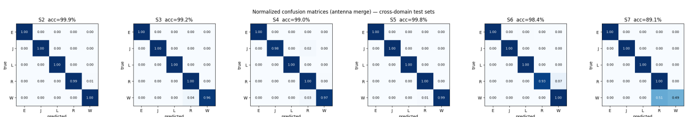

Fig. 5: Confusion matrices on the Cross-domain test set

In conclusion the mean merged accuracy across all domain test sets was 97.56%, validating the choice of Supervised Contrastive Learning as a feature extractor that can effectively clusters activity while remaining robust to domain-specific noise.

VII. Concluding Remarks

In this work, we successfully extended the capabilities of passive Wi-Fi sensing for Human Activity Recognition (HAR) and Person Identification (PI) by integrating contrastive learning paradigms into the SHARP framework. Rather than relying on standard cross-entropy classification, we demonstrated that mapping Doppler traces into a structured latent space-either through self-supervised multi-antenna views or label driven supervised contrastive learning-yields highly robust representations. Both pipelines achieved an outstanding accuracy.

Because our models are trained to focus on the underlying physical similarities of an activity rather than environment-specific noise, they can be deployed in novel physical layouts without requiring extensive retraining. This makes our approach highly applicable to real-world, privacy-preserving smart environments—such as elderly care facilities or personalized smart homes—where deploying cameras is intrusive, but adapting to new room layouts and users is essential.

Despite the high accuracy, our current methodology relies on a controlled dataset where activities are performed sequentially by a single subject at a time. A critical missing piece for real-world deployment is the handling of multi-subject, concurrent scenarios—for instance, identifying one person walking while another is sitting in the same room. Future iterations of this work should focus on source separation algorithms for overlapping Doppler signatures.

A. Project reflections

From a pedagogical and developmental perspective, this project provided profound insights into the mechanics of modern Deep Learning. We learned firsthand the strength of contrastive approaches: by shifting the objective from simply "predicting a label" to "understanding the structural similarity between samples" a model tends to learn more general features. Furthermore, working with the existing SHARP framework taught us the value of incremental research. We learned how to analyze an already well-built, state-of-the-art system, identify its bottlenecks, and modularly introduce new paradigms to expand its capabilities.

The development process was not without significant hurdles, particularly regarding the training dynamics. Contrastive learning frameworks, notably the NT-Xent loss function, are notoriously sensitive to hyperparameter configurations, such as batch size and the temperature scaling parameter. Finding a stable training strategy required extensive empirical testing. Moreover, designing the data augmentation pipeline for the self-supervised approach proved challenging. Unlike images, where crops or color distortions are easily verified by the human eye, applying random transformations (like temporal masking or patch erasing) to complex Doppler spectrograms ran the risk of destroying the underlying physical meaning of the signal. Balancing aggressive augmentation—needed for robust contrastive learning—with signal integrity was one of the most demanding, yet rewarding, aspects of this project.

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